Consciousness and Cognition 22 (2013) 875–886

Contents lists available at SciVerse ScienceDirect

Consciousness and Cognition

journal homepage: www.elsevier.com/locate/concog

Conscious recollection and binding among context features

C. Dennis Boywitt *, Thorsten Meiser

School of Social Sciences, Department of Psychology, University of Mannheim, Germany

ARTICLE INFO

Article history:
Received 19 February 2013
Available online 19 June 2013

Keywords:
Conscious recollection
Remember/know
Episodic memory
Source memory
Context binding

ABSTRACT

Recent research suggests that the subjective feeling of conscious recollection is uniquely characterized by joint memory for several context features while merely familiar memories lack this property (Meiser, Sattler, & Weisser, 2008). In the present research we took the novel approach of extending the dual task paradigm to the simultaneous study of subjective retrieval experience (using the remember/know procedure) and joint memory for two orthogonal context features. While dual task load during encoding lead to reductions in the frequency of the subjective experience of conscious recollection and reductions in overall context memory, joint context memory was *not* affected. Furthermore, the relation of higher overall context memory for consciously recollected items than for familiar items was preserved even under dual task load. These results have import implications for theories of long-term feature binding and the processes involved in producing the experience of conscious recollection.

1. Introduction

Episodic memories are defined by the retrieval of contextual details specifying the circumstances under which these events were encoded and the associated subjective experience has been termed conscious recollection. Memories which are accompanied by a mere feeling of familiarity, on the other hand, are not thought to be associated with the retrieval of episodic details (Tulving, 1985). While there is empirical evidence that conscious recollection is indeed associated with better memory for contextual details than is a feeling of familiarity (Dewhurst & Hitch, 1999; Dudukovic & Knowlton, 2006; Meiser & Bröder, 2002; Perfect, Mayes, Downes, & VanEijk, 1996), recent research has demonstrated that conscious recollection is more specifically characterized by joint memory for different context details from the encoding episode (Boywitt & Meiser, 2012a, 2012b; Meiser & Bröder, 2002; Meiser et al., 2008).

These recent studies show a consistent relationship between the subjective retrieval experience (as assessed with the remember/know paradigm; Tulving, 1985) and joint retrieval of the context details such that ‘remember’ responses were associated with joint memory of the context features whereas ‘know’ responses were not. The process underlying joint memory of different context features was termed context–context binding (Meiser & Bröder, 2002). As opposed to simple item-context binding, context–context binding may be conceived as a stronger concatenation of the context features in the memory trace, increasing the probability of joint recall of context features (Boywitt & Meiser, 2012a). In line with the conceptualization of episodic memories as integrated representations of the study episode (e.g., Parkin & Walter, 1992; Perfect et al., 1996; Rajaram, 1993; Tulving, 1985) these studies suggest that memories which are experienced as consciously recollected are coherent re-instantiations of the feature configuration at study while memories which feel only familiar lack this property. The present research pursued two interrelated goals. First, we aimed to fill the gap in knowledge of the

* Corresponding author. Address: Department of Psychology, University of Mannheim, D-68131 Mannheim, Germany. Fax: +49 621 181 2129.
E-mail address: boywitt@uni-mannheim.de (C.D. Boywitt).

1053-8100/$ - see front matter © 2013 Elsevier Inc. All rights reserved.
http://dx.doi.org/10.1016/j.concog.2013.05.010

underlying processes involved in creating context–context binding as a correlate of recollective experience. Second, we sought to shed light on the relation of conscious recollection and context memory in general.

2. Mechanisms of context–context binding

Previous research suggests that context–context binding is produced by an encoding process operating in the immediate focus of attention which integrates the features of the episode into a coherent memory representation (Boywitt & Meiser, 2012b; Uncapher, Otten, & Rugg, 2006). Evidence for this view comes from an imaging study indicating that the intra-parietal sulcus, which is thought to be involved in attentionally mediated binding, is especially activated during encoding of items with two independent context features which are later remembered with both context features as opposed to items which are later remembered with only one context feature (Uncapher et al., 2006). Uncapher et al. suggested that allocation of attention to the "object level" increases the likelihood of establishing bound representations. Behavioral studies offer converging evidence that context–context binding is only observed for context features which are in the focus of processing during encoding (Boywitt & Meiser, 2012b). In an incidental learning situation, context–context binding was only observed for stimulus-bound (i.e., intrinsic) context features (i.e., font color) but not for more peripheral (i.e., extrinsic) context features (i.e., color of a frame), which are not in the immediate focus of attentional processing (Ecker, Zimmer, & Groh-Bordin, 2007; Mulligan, 2011). When intrinsic and extrinsic context features were in the focus of processing due to explicit learning instructions, however, context–context binding was observed for both intrinsic and extrinsic context features, suggesting that context–context binding depends on an attentionally mediated encoding mechanism.

Further evidence comes from research on visual short-term memory, suggesting that binding of different perceptual features is mediated by a resource-demanding process (e.g., Brown & Brockmole, 2010; Rensink, 2000; Stefurak & Boynton, 1986; Treisman & Gelade, 1980; Treisman & Schmidt, 1982; Wheeler & Treisman, 2002). To illustrate, in a visual short-term memory task, Wheeler and Treisman (2002) presented participants with multiple objects simultaneously, which differed in location and color. While change detection performance for the independent features was quite good, changes that affected the binding between location and color such as re-arranged pairs of previously presented features were detected at a much lower rate. Additional empirical evidence for a resource-demanding binding process comes from a study in which participants were required to perform a mental arithmetic task during the retention interval between the presentation of colored shapes and the test phase (Stefurak & Boynton, 1986). This manipulation affected the detection of new shape-color pairings while accuracy for the individual features was largely preserved. In a similar vein, Treisman and Schmidt (1982) have shown that the rate of conjunction errors (i.e., false recognition of re-arranged features from different objects) increased under cognitive load during presentation of the stimuli. Similarly, another study found verbal–spatial binding to be impaired by a concurrent working memory load during stimulus presentation (Elsley & Parmentier, 2009), indicating that the integration of multiple features relies on a resource-demanding process.

Although there is ample evidence that feature binding may rely on resource-demanding processes, there is also evidence suggesting that binding of different features might not be resource-demanding over and above the demands of encoding and maintaining the individual features (e.g., Allen, Baddeley, & Hitch, 2006; Delvenne, Cleeremans, & Laloyaux, 2010; Luck & Vogel, 1997). Allen et al. (2006), for example, did not find decreased recognition performance in a short-term memory task for combinations of features under additional load as compared with the features alone.

3. Goals of the present research

Because previous research has provided only indirect evidence on the attentional demands of context–context binding, we tested the prediction that binding of context features depends on resource-demanding processes (Boywitt & Meiser, 2012a, 2012b; Uncapher et al., 2006) directly by implementing a dual-task manipulation. For that purpose, we employed a multidimensional source memory paradigm with two completely crossed context dimensions (Meiser & Bröder, 2002) and assessed subjective retrieval experience with the remember/know-procedure. In the multidimensional source memory paradigm participants study items varying on two orthogonal context dimensions, for example items presented at two different screen positions and two different font sizes. Assessing memory for two completely crossed context dimensions allows for the analysis of stochastic dependence in context memory. For instance, if retrieving one context feature (e.g., font size) is more likely if the other context feature (e.g., screen position) is also retrieved compared with retrieving the feature font size alone, then context memory is stochastically dependent. The phenomenon of stochastic dependence in memory for context features has been taken as an indication of context–context binding.

Theoretically, stochastic dependency in memory for two orthogonal context features is independent from "simple" memory for each context feature. That is, memory could be fairly accurate for both context features but may still be stochastically independent (cf. Light & Berger, 1976). Indeed, this pattern of results has been observed when "simple" context memory was experimentally equated between *R* and *K* judgments but stochastic dependency was still only observed for *R* responses, suggesting a qualitative difference in the underlying memory characteristics of *R* and *K* responses (Meiser et al., 2008).

In the first experiment stochastic dependency in context memory as an indication of context binding was compared between a condition with dual task (DT) load during study and a full attention condition. In the second experiment we broadened our focus to include the relation of subjective retrieval experience and context memory in general.

4. Experiment 1

If, as much of the literature on feature binding suggests, context–context binding also relies on a resource-demanding encoding process then DT load during encoding should diminish context–context binding. Alternatively, context–context binding might be spared by DT load during encoding, suggesting that the underlying process is rather automatic and/or occurs during retrieval.

Complicating the matter, previous research suggests that DT load reduces overall context memory (e.g., Craik, Luo, & Sakuta, 2010; Troyer, Winocur, Craik, & Moscovitch, 1999) and thus, a reduction in context–context binding under DT load would be confounded with lower overall context memory. In order to avoid such a confound we equated overall context memory between the DT group and the control group by making the context features more difficult to distinguish for the control group. More similar context features have previously been shown to impair context memory (e.g., Bayen, Murnane, & Erdfelder, 1996), so that the manipulation is suitable to equate overall context memory across groups.

4.1. Method

4.1.1. Participants

Ninety-five students (mean age = 22.6, *SD* = 4.29 years; 71.6% female) participated in this study for either course credit or monetary compensation and were randomly assigned to the two groups. Forty-nine participants were assigned to the control group and 46 participants were assigned to the DT group.

4.1.2. Materials and design

From a set of 165 concrete German nouns of four to seven letters, 64 words were randomly drawn as study items and 40 words were randomly drawn as distracters for the test phase for each participant anew.

Because the dimensions font size (large vs. small) and position on the screen (left vs. right) were used in several previous studies (e.g., Boywitt, Kuhlmann, & Meiser, 2012; Meiser & Bröder, 2002; Meiser et al., 2008; Riefer, Chien, & Reimer, 2007; Starns & Hicks, 2008) we used the same context dimensions in order to be consistent with previous research and to allow comparisons with previous results. In order to equate context memory across groups, the context features were more similar for participants in the control group than for participants in the DT group. During the study phase, half of the 64 study items appeared on the left side of the screen and the other half appeared on the right side of the screen. Orthogonal to the position on the screen, the words varied with regard to font size. For participants in the DT group, the distance between the centers of item presentation was 19 cm whereas for participants in the control group the distance was only 15.2 cm. For participants in the DT group, the larger font size was 2.2 cm letter height and the smaller was 0.9 cm letter height. For participants in the control group, the letter heights were 1.5 cm and 0.9 cm, respectively. To balance the appearance of both features across the study list, the list was divided into four blocks of 16 items. In each block, each combination of the features location and font size appeared equally often in an order that was randomized for each participant anew. Presentation duration was three seconds with an ISI of one second in both groups.

For the secondary task a tone monitoring task was employed for which participants were presented with a random sequence of two tones differing in pitch (440 Hz and 330 Hz). Each tone lasted for one second followed by an ISI of two seconds. The participants were instructed to press the space bar on the keyboard whenever they heard three consecutive tones of the same pitch. This task should tap central executive resources without interfering with the visual modality in which the stimuli were presented (Gardiner & Parkin, 1990; Parkin & Russo, 1990).

4.1.3. Procedure

Instructions, presentation of materials, and the memory test were computer directed. The initial instructions were identical for the group with the tone monitoring task and the control group and informed participants that they would be presented with a series of words which were to be memorized for a later memory test. Importantly, participants were not instructed to memorize the font size and location of the words. Participants in the group with the tone monitoring task were then informed about the additional task and were given a short practice block of 8 tones. Following the instructions, both groups were presented the 64 study items one at a time. After the study phase, all participants were asked to write down as many European capitals as possible for three minutes. Subsequent to this filler task all participants received instructions concerning the memory test. Then the combined old/new, remember/know, context memory test commenced. During the test phase, target items and distracters were displayed in random order in the center of the screen in a font size intermediate between the large font size and small font size used in the study phase. All responses to the memory test were made by clicking with the mouse on response fields on the screen. For each item, participants were first asked to indicate whether the word was "old" or "new". If participants selected "old", they were asked to indicate whether they "remember" or "know" the word. Then, participants were asked to indicate in which combination of font size and position on the screen they had studied the word. For that response, four response fields with the four combinations of font size and location were displayed. The instructions for the remember/know judgments were identical to the instructions used in previous studies (Boywitt & Meiser, 2012b; Meiser & Bröder, 2002) and avoided any explicit association of *R* responses with the recollection of perceptual details like font size or location.

4.2. Results

An alpha level of .05 was specified for all analyses.

4.2.1. Remember-know judgments

Table 1 shows the proportion of *R* and *K* responses to target items (i.e., hits) and distracters (i.e., false alarms).

A two by two mixed ANOVA with type of judgment (*R* vs. *K*) as within participants factor and group (DT vs. control) as between participants factor on hits revealed a significant main effect of type of judgment, $F(1,93) = 23.60$, $MSE = 0.05$, $\eta_p^2 = .20$, $p < .001$, with higher rates of hits accompanied by *R* than *K* judgments. The main effect of group was also significant, $F(1,93) = 8.21$, $MSE = 0.02$, $\eta_p^2 = .08$, $p = .005$, indicating lower hit rates for the DT group than the control group. The interaction was marginally significant, $F(1,93) = 3.37$, $MSE = 0.05$, $\eta_p^2 = .04$, $p = .07$.

Independent *t*-tests revealed that the rate of *R* responses was lower in the DT group than in the control group, $t(93) = 2.56$, $d = .52$, $p = .012$, but rates of *K* responses did not differ between groups, $t(93) = 0.17$, $d = 0.03$, $p = .887$.

The same two by two mixed ANOVA on false alarms revealed a significant main effect of type of judgment, $F(1,93) = 53.30$, $MSE = 0.01$, $\eta_p^2 = .36$, $p < .001$, with higher false alarm rates for *K* responses than for *R* responses. Neither the main effect of group nor the interaction was significant, $F(1,93) = 1.70$, $MSE = .01$, $\eta_p^2 = .02$, $p = .20$, and $F(1,93) = 0.42$, $MSE = 0.01$, $\eta_p^2 = .004$, $p = .52$, respectively.

4.2.2. Overall context memory

Next we tested whether overall context memory differed between the two groups. To measure overall context memory, we computed for all correctly recognized target items the probability of assigning the items to their respective context for each context dimension (i.e., font size and location) separately. Table 2 displays the average conditional source-identification measure (ACSIM; Murnane & Bayen, 1996) scores. One-sample *t*-tests revealed that context memory was above chance level (i.e., >.5) for *R* responses in both groups (see Table 2).

A mixed ANOVA on ACSIM scores with the within participants factors context dimension (position vs. font size) and judgment (*R* vs. *K*) and the between participants factor group (control vs. DT) revealed a main effect of judgment, $F(1,88) = 105.75$, $MSE = 0.03$, $\eta_p^2 = .55$, $p < .001$, with higher context memory for *R* than for *K* responses. The main effect of context dimension was marginally significant, $F(1,88) = 3.90$, $MSE = 0.02$, $\eta_p^2 = .04$, $p = .051$, with slightly higher context memory for position than for font size. Neither the main effect of group was significant, $F(1,88) = 0.21$, $MSE = 0.04$, $\eta_p^2 = .002$, $p = .65$ nor any of the interactions: Judgment by group, $F(1,88) = 1.17$, $MSE = 0.03$, $\eta_p^2 = .01$, $p = .282$, Judgment by context dimension, $F(1,88) = 2.87$, $MSE = 0.02$, $\eta_p^2 = .03$, $p = .094$, context dimension by group, $F(1,88) = 0.69$, $MSE = 0.02$, $\eta_p^2 = .01$, $p = .407$, three-way interaction, $F(1,88) = 0.02$, $MSE = 0.02$, $\eta_p^2 < 0.001$, $p = .904$. Thus, context memory was successfully equated between both groups.

4.2.3. Model-based analyses

In the next step we analyzed the responses in the memory test with a re-parameterized variant of the multinomial processing tree model for two crossed source dimensions (Meiser, 2009; Meiser & Bröder, 2002). The model-based analyses offer

Table 1
Mean proportions (and standard errors) of remember and know responses to target items (hits) and distracters (false alarms) in Experiments 1 and 2.

	Experiment 1		Experiment 2		
	Control	Dual task at encoding	Control	Dual task at encoding	Dual task at retrieval
Hits					
Remember	.47 (.03)	.35 (.03)	.51 (.03)	.33 (.03)	.52 (.03)
Know	.24 (.02)	.25 (.02)	.21 (.02)	.27 (.02)	.24 (.02)
False alarms					
Remember	.03 (.01)	.04 (.01)	.02 (.01)	.04 (.01)	.02 (.004)
Know	.11 (.02)	.14 (.02)	.05 (.01)	.14 (.01)	.08 (.01)

Table 2
Mean conditional source identification (ACSIM) scores (and standard errors) in Experiments 1 and 2.

	Experiment 1		Experiment 2		
	Control	Dual task at encoding	Control	Dual task at encoding	Dual task at retrieval
Remember					
Position	.63 (.02)[a]	.65 (.03)[a]	.72 (.02)[a]	.65 (.02)[a]	.71 (.02)[a]
Font	.59 (.02)[a]	.59 (.02)[a]	.64 (.02)[a]	. 64 (.02)[a]	.60 (.02)[a]
Know					
Position	.45 (.02)	.43 (.03)	.63 (.03)[a]	.50 (.02)	.57 (.02)[a]
Font	.46 (.03)	.42 (.02)	.54 (.02)	.56 (.02)[a]	.55 (.02)[a]

[a] Value significantly above .5 ($p < .05$).

a variety of advantages over traditional analyses. First, these analyses allow for the simultaneous estimation of latent parameters for recognition memory, the subjective feelings of conscious recollection and familiarity, source memory, response biases, and guessing processes. Second, the model allows for the estimation of latent parameters indicating independent memory for each source dimension and for the analysis of stochastic dependency in source memory. The parameter estimates are frequently preferred over behavioral measures which may be contaminated by nuisance processes such as guessing and response tendencies (Batchelder & Riefer, 1990; Bayen et al., 1996; Erdfelder et al., 2009).

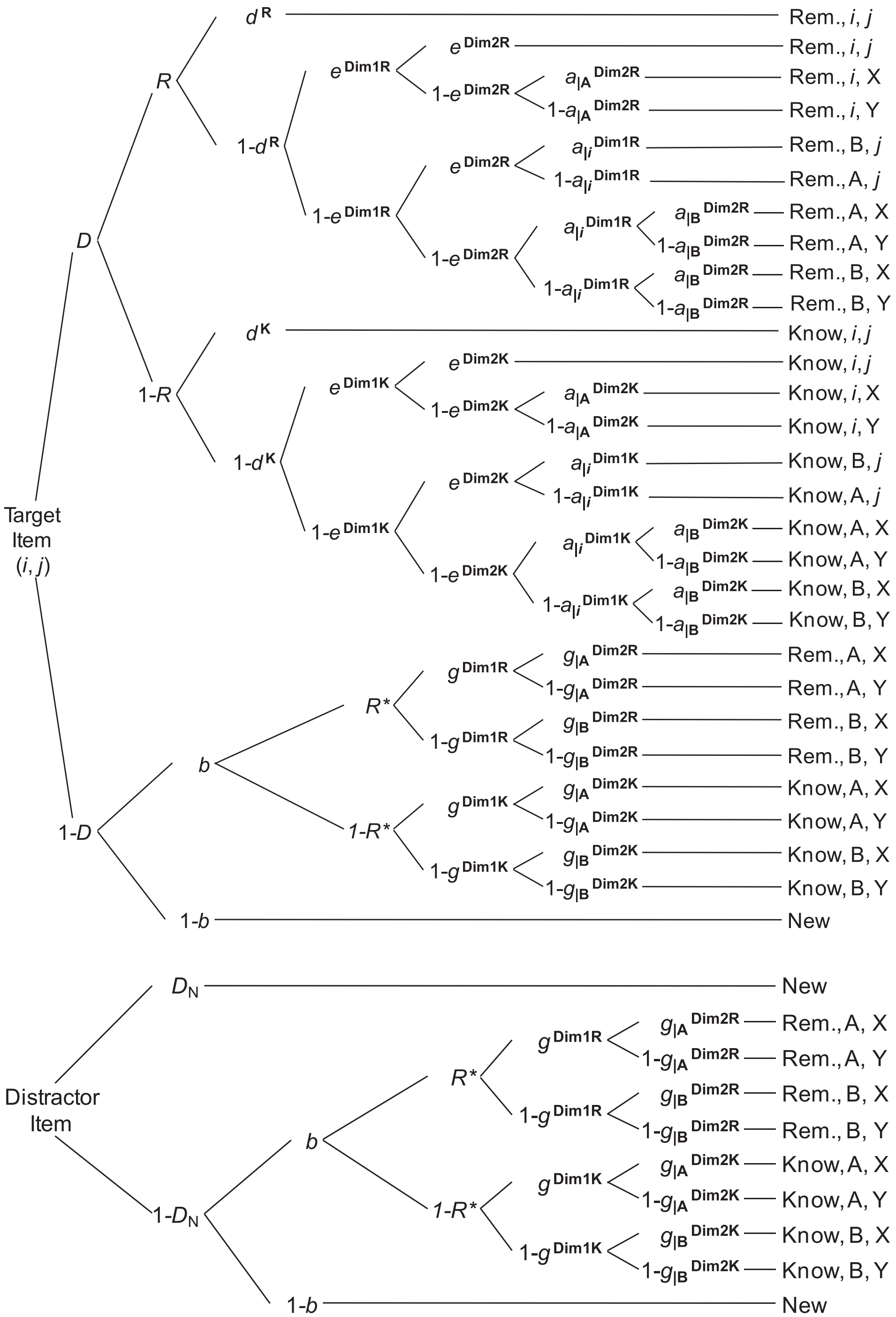

Fig. 1. Processing-tree diagram of the re-parameterized variant of the multinomial model of multidimensional source memory (Meiser, 2009; Meiser & Bröder, 2002), taken from "The role of source memory in older adults' recollective experience" by C.D. Boywitt, B.G. Kuhlmann, and T. Meiser (2012), Psychology and Aging, 27, 484–497. Adapted with permission.

Consider Fig. 1 for the following explanation of the models' parameter. The stimuli on the left side of Fig. 1 are connected with the observed response on the right side via the latent cognitive states represented by the model parameters. Given that the participant is presented with a previously studied item (i.e., "target item") at test (upper tree in Fig. 1), the parameter D denotes the probability of recognizing this target item as "old". The parameter R represents the probability that this recognized target item is judged as "remembered" whereas the complementary probability 1 - R represents the case when a recognized target item is judged as "known". Thus, this parameter indicates the propensity with which recognized items are experienced as "remembered" controlling for item recognition performance (parameter D) and response biases (parameters b and R^*). Accordingly, all subsequent parameters are specified conditional on the "remember"/"know" distinction as indicated with the superscript R or K. Given that participants do not recognize a target item as old ($1-D$) they may still guess that this item is old (b). The parameter b thus denotes the probability of guessing that an unrecognized old item or an undetected distracter is old. In this case participants may then erroneously give a R response with probability R^* and a K response with probability $1-R^*$.

The parameters d^R and d^K represent the probabilities of joint memory for both source dimensions over and above the contribution of independent source memory (parameters $e^{\mathrm{Dim}1R}$, $e^{\mathrm{Dim}2R}$, and $e^{\mathrm{Dim}1K}$, and $e^{\mathrm{Dim}2K}$). Accordingly, if $d^R > 0$ or $d^K > 0$ then memory for both context features is stochastically dependent indicating context–context binding. Thus, unlike the analyses of the behavioral source memory measure (ACSIM), the model-based analyses allow us to specifically measure context binding for R and K responses in terms of stochastic dependence in context memory. If d^R is smaller in the DT group than in the control group, this suggests that context–context binding was reduced by DT load over and above any effects on R and D. If d^R does not differ between the DT and the control group, on the other hand, context–context binding was not dissociated from R responses.

Last, the guessing processes in source memory decisions when source-specifying features cannot be remembered are captured by the parameters $a^{\mathrm{Dim}1R}_{|A/B}$, $a^{\mathrm{Dim}2R}_{|A/B}$, $a^{\mathrm{Dim}1K}_{|A/B}$, $a^{\mathrm{Dim}2K}_{|A/B}$ given a studied item was classified as R or K, respectively. The parameters $g^{\mathrm{Dim}1R}_{|A/B}$, $g^{\mathrm{Dim}2R}_{|A/B}$, $g^{\mathrm{Dim}1K}_{|A/B}$, and $g^{\mathrm{Dim}2K}_{|A/B}$ capture the guessing processes in source attributions if a study item was not detected as old ($1-D_N$) but was still guessed to be old (with probability b).

Given the participant is presented with a new item at test (i.e., "distractor item" in the lower tree in Fig. 1) the parameter D_N denotes the probability of identifying this distracter item as "new". Given that participants do not detect a distracter item (with probability $1-D_N$) and erroneously judge this item as "old" (with probability b) they may still guess that they "remember" the item with probability R^* or that they "know" the item with the complementary probability $1-R^*$. In both cases the parameters $g^{\mathrm{Dim}1R}_{|A/B}$, $g^{\mathrm{Dim}2R}_{|A/B}$, $g^{\mathrm{Dim}1K}_{|A/B}$, and $g^{\mathrm{Dim}2K}_{|A/B}$ capture the guessing processes in source attributions.

All model-based analyses were conducted with data aggregated at the group level using the computer program multiTree (Moshagen, 2010). Based on previous studies (Meiser & Bröder, 2002) we expected item memory for words presented in large font to be superior to item memory for words presented in small font and therefore separate item recognition parameters (D) were estimated for the two font size conditions. In order to render the model identifiable, distracter detection (parameter D_N) was set equal to target recognition (parameter D) of the small font size condition as is common in two-high threshold models of recognition memory (e.g., Meiser & Bröder, 2002).

The model provided a good fit to the data in the control group, $G^2(17) = 21.96$, $p = .19$, and in the group with DT, $G^2(17) = 16.89$, $p = .46$. Table 3 displays the parameter estimates.

Item recognition performance (D parameters) in the DT group was generally lower than in the control group, $G^2(2) = 84.43$, $p < .001$. Furthermore, neither the probability of assigning R responses to recognized items (parameter R) differed between the control group and the DT group, $G^2(1) = 1.77$, $p = .184$, nor the probability of assigning R responses to erroneously endorsed distracters and undetected old items (parameter R^*), $G^2(1) = 0.56$, $p = .452$.

Table 3
Parameter values (and 95% confidence intervals) estimated with the multinomial model of multidimensional source memory in Experiments 1 and 2.

Parameter	Experiment 1		Experiment 2		
	Control	Dual task at encoding	Control	Dual task at encoding	Dual task at retrieval
$D^{\mathrm{smallfont}}$	.56 (.53;.59)	.40 (.37;.43)	.63 (.60;.65)	.40 (.37;.43)	.65 (.62;.67)
$D^{\mathrm{largefont}}$	.61 (.57;.64)	.47 (.43;.51)	.69 (.66;.71)	.47 (.44;.51)	.70 (.67;.73)
R	.76 (.73;.80)	.73 (.69;.77)	.74 (.72;.77)	.68 (.65;.72)	.74 (.72;77)
R^*	.17 (.13;.22)	.20 (.15;.24)	.30 (.23;.38)	.21 (.17;.26)	.24 (.18;.30)
b	.31 (.28;.33)	.28 (.26;.31)	.18 (.15;.20)	.29 (.26;.31)	.27 (.24;.30)
d^R	.12 (.05;.19)	.15 (.05;.24)	.20 (.13;.27)	.14 (.06;.23)	.15 (.09;.21)
d^K	.00 (.00;.14)[a]	.02 (.00;.16)[a]	.06 (.00;.16)[a]	.00 (.00;.15)[a]	.10 (.00;.20)[a]
$e^{\mathrm{position}R}$	.09 (.00;.18)	.09 (.00;.22)[a]	.10 (.01;.20)	.18 (.07;.29)	.07 (.00;.15)[a]
$e^{\mathrm{position}K}$	.10 (.00;.28)[a]	.00 (.00;.21)[a]	.09 (.00;.22)[a]	.19 (.02;.37)	.04 (.00;.19)[a]
$e^{\mathrm{fontsize}R}$	.17 (.09;.26)	.30 (.20;.41)	.30 (.23;.38)	.19 (.08;.29)	.25 (.18;.33)
$e^{\mathrm{fontsize}K}$	.16 (.00;.33)[a]	.08 (.00;.29)[a]	.24 (.11;.37)	.08 (.00;.27)[a]	.12 (.00;.26)[a]

Note. D = probability of recognizing a study word; R = probability of assigning a R response to a recognized studied word; R^* = probability of assigning a R response to distracter words that were guessed to be old; b = probability of classifying an undetected distracter as "old"; $d\,R$ = probability of joint context memory for study words classified as R; $d\,^K$ = probability of joint context memory for study words classified as K; e = independent memory for the context features, superscript indicates classification of memory experience.

[a] Confidence intervals with limits of .00 were bounded by the parameter space which limits the range of each model parameter to the interval (.00, 1.00).

In the next step, we compared the parameters d^R and d^K across groups to assess whether context–context binding was affected by DT load. There was no overall effect of DT on the binding parameters for R and K responses, $G^2(2) = 0.33$, $p = .847$, but the binding parameters were generally larger for R than for K responses, $G^2(2) = 8.21$, $p = .017$. Furthermore, the confidence intervals for the binding parameter for K responses in both groups overlapped with zero but they did not overlap with zero for R responses, indicating that context binding was absent in K responses and present in R responses in both groups.

In order to ensure that the lack of a difference in the binding parameter (d^R) between the control group and the DT group is not due to a lack of statistical power, we conducted a power analysis using the computer program multiTree (Moshagen, 2010). The results indicate that the power for detecting an absolute difference of .20 (effect size $w \approx .04$) in the parameter estimates with the given sample ($N = 9880$ [104 items $\times$ 95 participants]) was $1 - \beta = .94$. Thus, the statistical power was sufficient to detect substantial effects of the DT manipulation on context–context binding.

Last, comparing independent context memory for both dimensions and R and K responses (e^{Dim1R}, e^{Dim1K}, e^{Dim2R}, and e^{Dim2K}) between both groups revealed no significant overall differences in independent context memory, $G^2(4) = 6.92$, $p = .140$.

4.3. Discussion

The results of the analysis of R and K responses are in line with previous studies which employed DT manipulations and elicited *RK* judgments: DT load selectively reduced the rate of R responses to target items and left the rate of K responses unaffected (Gardiner, Gregg, Mashru, & Thaman, 2001; Gardiner & Parkin, 1990). Further, overall context memory (as measured via ACSIM scores) was higher for R than for K responses in both groups, converging with results of other studies (e.g., Dudukovic & Knowlton, 2006; Meiser & Bröder, 2002). Importantly, the similarity manipulation of the context features was successful, as indicated by similar levels of context memory in the DT and the control group.

The model based analyses, however, suggest a different conclusion regarding the effect of DT load on *RK*-judgments. While the analysis of the raw frequencies of R and K responses indicates a lower rate of R responses in the DT group than in the control group, the model-based analysis did not show any difference in the probability of assigning R responses to recognized items (parameter R). Note, however, that the raw frequencies of R and K responses are two-in-one measures of item recognition *and* subjective retrieval experience because the observed rates of R and K responses add up to the total rate of "old" responses. Accordingly, manipulations which affect item recognition performance or subjective retrieval experience both manifest themselves in differences in R and K responses. The model-based analysis, on the other hand, decomposes the observed frequency of R and K responses into the contributions of item recognition (parameter D), subjective retrieval experience (parameter R), and guessing processes (parameters b and R^*). Thus, according to the model-based analysis the observed difference in the rate of R responses was driven by differences in mere item recognition performance (parameter D) between the control group and the DT group and not by differences in genuine recollective experience (parameter R). Assuming that the recollective experience is associated with retrieval of some context details (Yonelinas, 2002) this result can be interpreted as a consequence of the equating of context memory between the two groups: Because there were no differences in context memory between the DT and the control group, no differences in recollective experience could emerge.

Last but most importantly, the typical pattern of context–context binding for R but not for K responses was observed in the control group *and* in the group with the DT load during encoding. This result suggests that context–context binding was equally prone to reductions in available resources during encoding as was conscious recollection. Thus, DT load did not alter the relation between context–context binding and conscious recollection.

5. Experiment 2

The central result of the first experiment is that context–context binding was not reduced by cognitive load during encoding over and above the effects on subjective retrieval experience. This leaves open the question whether a retrieval process as opposed to an encoding process might be involved in joint context memory (Meiser & Bröder, 2002; Starns & Hicks, 2008). For example, a cueing process between the context features (Meiser & Bröder, 2002) or between the item itself and each context feature (Starns & Hicks, 2008) during retrieval might produce stochastic dependency in context memory. The notion that an additional (retrieval) process is involved in R responses is also compatible with extant dual process models of human memory (e.g., Jacoby, 1991; Mandler, 1980; Yonelinas, 1994). Most of these models assume that two different processes operate at time of retrieval, a familiarity process on the one hand, and a recollection process on the other hand (see Yonelinas, 2002, for a review) and this latter process might also be involved in producing joint context memory. To test the role of effortful retrieval processes in producing context–context binding, we extended the design of Experiment 1 by including a DT group during retrieval. Second, we sought to look into the relation of resource-demanding processes in the subjective retrieval experience and context memory in general which could not be investigated in the first experiment because context memory was experimentally equated between groups. Although many previous studies have assessed the effects of dual task load on *RK* judgments (Gardiner & Parkin, 1990; Gardiner et al., 2001; Yonelinas, 2001) *or* source memory (Craik et al., 2010; Troyer & Craik, 2000; Troyer et al., 1999) ours is the first study to our knowledge which jointly looks at the effects of dual task load on context memory for two orthogonal features *and RK* judgments. The parallel pattern of decreased rates of *R* responses and decreased source memory performance in those previous studies suggests similar effects for both source memory and recollective experience. What those studies do not reveal, however, is whether the *relation* of higher source memory for *R* than for *K* judgment is invariant across cognitive load conditions. Put more bluntly, is context memory still characteristic of *R* responses even under cognitive load? Indeed, there is evidence showing that this relationship may not be as universal as suspected (Hicks, Marsh, & Ritschel, 2002; Meiser & Sattler, 2007).

5.1. Method

5.1.1. Participants

One hundred and sixty-nine students (mean age = 20.97, SD = 2.16 years; 80% female) participated in this study for course credit or monetary compensation and were randomly assigned to the three groups. Fifty-five participants were assigned to the control group, 57 participants each were assigned to the dual task during encoding group and to the dual task during retrieval group.

5.1.2. Materials and design

The materials were identical to the materials used in the first experiment with the following exceptions. The distance between the left and right position on the screen was 19 cm between the centers of the stimuli in all groups. The large font size was 78 pt font and the smaller font size was 34 pt font in all groups.

The control group performed both the study and the test phase without a secondary task and one group received the additional tone monitoring task during the study phase while another group received the tone monitoring task during the test phase.

5.1.3. Procedure

The procedure was identical to the procedure in the first experiment with the following exceptions. Participants in the group with the tone monitoring task during retrieval were informed about the additional task prior to the memory test and were given the same short practice block of the 8 tones as the group with the tone-monitoring task during encoding. The test block was divided into two parts of equal length with a break of one minute in between for all participants to avoid motor fatigue effects in the group with the tone monitoring task during the test phase.

5.2. Results

5.2.1. Remember-know judgments

Table 1 shows the proportions of *R* and *K* responses to target items (i.e., hits) and distracters (i.e., false alarms). Concerning hits, a two by three mixed ANOVA with type of judgment (*R* vs. *K*) as within participants factor and DT group (control vs. load during encoding vs. load during retrieval) as between participants factor revealed a significant main effect of type of judgment, $F(1,166) = 76.12$, $MSE = 0.051$, $\eta_p^2 = 0.31$, $p < .001$, with higher hit rates for *R* than for *K* judgments. The main effect of DT group was also significant, $F(2,166) = 13.70$, $MSE = 0.015$, $\eta_p^2 = 0.14$, $p < .001$. However, the main effects were qualified by a significant interaction, $F(2,166) = 8.98$, $MSE = 0.051$, $\eta_p^2 = 0.10$, $p < .001$. Separate one-way ANOVAs on the hit rate with *R* responses and *K* responses revealed a main effect of DT group on the rate of *R* responses, $F(2,166) = 12.51$, $MSE = 0.050$, $\eta_p^2 = 0.13$, $p < .001$, but not on the rate of *K* responses, $F(2,166) = 2.53$, $MSE = 0.016$, $\eta_p^2 = 0.03$, $p = .083$. Post hoc tests (LSD) on the rate of *R* responses showed that the group with a secondary task during encoding produced significantly fewer *R* responses than the control group, $p < .001$, and the group with the secondary task during retrieval, $p < .001$, but the control group and the group with a secondary task during retrieval did not differ significantly, $p = .72$.

Concerning false alarms, a two by three mixed ANOVA with the factors judgment and DT group revealed a significant main effect of type of judgment, $F(1,166) = 67.36$, $MSE = 0.004$, $\eta_p^2 = 0.29$, $p < .001$, with higher false alarm rates for *K* than for *R* responses, and a main effect of DT group, $F(2,166) = 13.36$, $MSE = 0.006$, $\eta_p^2 = .14$, $p < .001$. The interaction was also significant, $F(2,166) = 8.95$, $MSE = 0.004$, $\eta_p^2 = 0.14$, $p < .001$. Separate one-way ANOVAs revealed that the groups did not differ in their false *R* rate, $F(2,166) = 1.84$, $MSE = 0.003$, $\eta_p^2 = 0.02$, $p = .16$, but differed in their false *K* rate, $F(2,166) = 14.57$, $MSE = 0.008$, $\eta_p^2 = 0.15$, $p < .001$. Post hoc tests (LSD) showed that this effect was due to the higher false *K* rate in the DT during encoding group than in both other groups, p's $< .001$, while false alarm rates in the control group did not differ from the false alarm *K* rate in the DT during retrieval group $p = .11$.

These results suggest that the DT during encoding selectively reduced the rate of *R* responses to study items and left the rate of *K* responses to study items unaffected. Similarly, false alarms increased only in the group with the DT during encoding selectively for *K* responses.

5.2.2. Overall context memory

Again, overall context memory was above chance level for *R* responses in all groups (see Table 2).

A mixed ANOVA on the ACSIM scores with the within participants factors context dimension (position vs. font size) and judgment (*R* vs. *K*) and the between participants factor group (control vs. DT at encoding vs. DT at retrieval) revealed a main effect of judgment, $F(1,150) = 74.50$, $MSE = 0.02$, $\eta_p^2 = .33$, $p < .001$, with higher context memory for *R* than for *K* responses.

The main effect of context dimension was also significant, $F(1,150) = 12.39$, $MSE = .02$, $\eta_p^2 = .08$, $p = 001$, with better context memory for position than for font size. The main effect of group was marginally significant, $F(2,150) = 2.98$, $MSE = 0.04$, $\eta_p^2 = .04$, $p = .054$. Neither the interaction between judgment and DT group was significant, $F(2,150) = 0.32$, $MSE = 0.02$, $\eta_p^2 = .004$, $p = .726$, nor the three-way interaction, $F(1,150) = 2.27$, $MSE = 0.02$, $\eta_p^2 = .03$, $p = .107$. The interaction between context dimension and judgment, however, was significant, $F(1,150) = 5.67$, $MSE = 0.02$, $\eta_p^2 = .04$, $p = .019$, as was the interaction between context dimension and DT group, $F(1,150) = 8.53$, $MSE = 0.02$, $\eta_p^2 = .10$, $p < .001$.

In order to probe these interactions, mixed ANOVAs with the within participants factor judgment (*R* vs. *K*) and the between participants factor group (control vs. DT at encoding vs. DT at retrieval) were conducted separately for ACSIM scores of font size and of position. For font size the analysis revealed a significant main effect of judgment, $F(1,150) = 28.96$, $MSE = 0.02$, $\eta_p^2 = .16$, $p < .001$, with higher context memory for *R* than for *K* responses but no main effect of DT group, $F(2,150) = 0.02$, $MSE = 0.03$, $\eta_p^2 = .008$, $p = .559$, and no interaction, $F(1,150) = 1.04$, $MSE = .02$, $\eta_p^2 = .01$, $p = .356$. The same ANOVA on ACSIM scores for position, however, revealed in addition to the main effect of judgment, $F(1,150) = 56.56$, $MSE = 0.02$, $\eta_p^2 = .27$, $p < .001$, with better context memory for *R* than for *K* judgments, a significant effect of DT group, $F(2,150) = 8.22$, $MSE = 0.03$, $\eta_p^2 = .10$, $p < .001$, but no interaction, $F(1,150) = 1.25$, $MSE = 0.02$, $\eta_p^2 = .02$, $p = .29$. Post hoc tests (LSD) showed that context memory in the DT at encoding group was lower than in the control group, $p < .001$, and lower than in the DT at retrieval group, $p = .019$, but the latter two groups did not differ significantly, $p = .111$.

5.2.3. Model-based analyses

The multinomial model in Fig. 1 provided an adequate fit to the data in the control group, $G^2(17) = 26.06$, $p = .07$, and a good fit in the group with the DT during encoding, $G^2(17) = 6.39$, $p = .99$, and the group with the DT during retrieval, $G^2(17) = 19.84$, $p = .28$. Refer to Table 3 for the parameter estimates.

The main effect of DT group on item recognition (*D* parameters) was significant, $G^2(4) = 348.77$, $p < .001$. This effect was driven by overall lower recognition performance in the DT during encoding group than in the control group, $G^2(2) = 235.25$, $p < .001$, and in the dual task during retrieval group, $G^2(2) = 285.08$, $p < .001$, while the latter two groups did not differ significantly, $G^2(2) = 1.98$, $p = .372$.

The overall test of the equivalence of the probability of assigning *R* responses to recognized target items (*R* parameter) was also significant, $G^2(2) = 7.94$, $p = .019$. Post hoc comparisons revealed no significant difference in the *R* parameter between the DT at retrieval group and the control group, $G^2(1) = 0.01$, $p = .94$, but lower estimates in the DT at encoding group than in the control group, $G^2(1) = 7.19$, $p = .007$, and the DT at retrieval group, $G^2(1) = 6.74$, $p = .009$.

The probability of assigning an *R* response to erroneously endorsed distracters or undetected study items (parameter R^*) did not differ significantly between the three groups, $G^2(2) = 4.34$, $p = .114$.

Regarding, context–context binding, there were neither overall differences in the binding parameter for *R* responses (d^R) between the three groups, $G^2(2) = 1.65$, $p = .438$, nor in the binding parameter for *K* responses (d^K), $G^2(2) = 1.74$, $p = .419$. However, the binding parameters for *R* responses were generally larger than the binding parameters for *K* responses, $G^2(3) = 9.96$, $p = .019$. Furthermore, the confidence intervals for the binding parameter for *K* responses but not for *R* responses overlapped with zero in all groups. Thus, context–context binding was again absent in *K* responses and present in *R* responses for all groups.

Comparing independent context memory for *R* and *K* responses for both context dimensions (parameters $e^{\mathrm{Dim1}R}$, $e^{\mathrm{Dim1}K}$, $e^{\mathrm{Dim2}R}$, and $e^{\mathrm{Dim2}K}$) across groups revealed significant differences, $G^2(8) = 19.27$, $p = .013$. Post hoc tests showed that independent context memory differed significantly between the DT during encoding group and the control group, $G^2(4) = 16.62$, $p = .002$, and the DT during retrieval group, $G^2(4) = 12.01$, $p = .017$. But the latter two groups did not differ in independent context memory, $G^2(4) = 2.39$, $p = .665$. Probing this main effect, we compared context memory for each context dimension and *R* and *K* responses separately between the dual task condition during encoding and both other conditions jointly. Due to decreased degrees of freedoms for these analyses none of the post hoc comparisons was statistically significant, $e^{\mathrm{position}K}$: $G^2(1) = 2.03$, $p = .154$, $e^{\mathrm{position}R}$: $G^2(1) = 2.18$, $p = .139$, $e^{\mathrm{fontsize}K}$, $G^2(1) = 2.11$, $p = .146$, $e^{\mathrm{fontsize}R}$, $G^2(1) = 2.19$, $p = .139$.

5.3. Discussion

To summarize, the DT load during encoding yielded a lower rate of *R* responses but left *K* responses unaffected while DT during retrieval had almost no effect on *RK* judgments. Again, this pattern of results is well in line with previous studies (e.g., Gardiner & Parkin, 1990; Gardiner et al., 2001; Yonelinas, 2001) and highlights the importance of encoding processes for the subsequent recollective experience. A novel finding in this regard is that the DT load during retrieval did not have any detrimental effects on either the rates of *R* or *K* responses.

The analyses of overall context memory (as measured via ACSIM scores) revealed a selective reduction of context memory in the DT during encoding group but not in the DT during retrieval group, in line with previous studies (e.g., Craik et al., 2010; Troyer et al., 1999). Importantly, the *relation* between context memory and subjective retrieval experience such that *R* responses are associated with higher context memory than *K* responses was preserved even under DT load. Furthermore, the selective context memory impairment due to dual task load during encoding was limited to the dimension position. This result is consistent with the idea that context dimensions may differ with regard to the processing demands such that stimulus-bound (i.e., intrinsic) context features are less prone to DT manipulations than spatio-temporal (i.e., extrinsic) context features (Moscovitch, 1992; Spencer & Raz, 1995; Troyer et al., 1999). According to this reasoning, position on the screen can be considered a spatial–temporal feature which requires more effort to encode than font size which can be considered a more stimulus-bound feature. Reducing the available resources during encoding thus affected the resource-demanding encoding of position but not the encoding of font size.

In line with the results of the analysis of the behavioral data, the model-based analyses revealed that item recognition (D parameter) as well as recollective experience (R parameter) were significantly reduced by the DT load during encoding but not by DT load during retrieval. Unlike in Experiment 1, the difference in the observed rate of R responses between the DT load during encoding group and the control group appeared to have been driven not only by differences in item recognition performance (D parameter) but also by differences in genuine recollective experience (R parameter). In combination with lower overall context memory in the DT load during encoding group this result suggests that decreasing context memory decreases recollective experience. Crucially, context–context binding was still present for R but not for K responses in all three groups and did not differ between the DT groups and the control group even though overall context memory was higher, at least for the dimension position, in the control group than in the DT during encoding group.

These results corroborate the conclusions drawn from the first experiment that context–context binding of perceptual features does not rely on resource-demanding processing over and above the demands of conscious recollection. Thus, given the experience of consciously recollected memories, the quality of associated context memory did not differ across the three groups although this experience occurred substantially less frequently if a cognitive load was imposed during encoding.

6. General discussion

A substantial body of research indicates that the subjective experience of conscious recollection is associated with substantial memory for the context details of the encoding episode (e.g., Dewhurst & Hitch, 1999; Dudukovic & Knowlton, 2006; Meiser & Bröder, 2002; Perfect et al., 1996) and, more importantly, with bound memory of these context features (Meiser & Bröder, 2002; Meiser & Sattler, 2007; Meiser et al., 2008). Recently, it has been suggested that context–context binding is accomplished by an encoding process operating in the focus of attentional processing which integrates the stimulus and its context features into a coherent memory representation of the feature configuration during study (Boywitt & Meiser, 2012b; Uncapher et al., 2006). In line with this suggestion, research on short-term memory feature binding offers ample evidence that the integration of multiple perceptual features is resource-demanding (e.g., Stefurak & Boynton, 1986; Treisman & Schmidt, 1982; Wheeler & Treisman, 2002). Contrary to this hypothesis, however, in the first experiment context–context binding was present in R responses even under DT load during encoding when overall context memory was equated between the DT group and the control group. The second experiment corroborated this finding and extended it to DT manipulations during retrieval. In both experiments context–context binding was correlated with the recollective experience which was only affected by DT load during encoding but not by DT load during retrieval. Thus, context–context binding appears to be associated with the processes driving conscious recollection but there was no evidence that effortful processing over and above the processing for recollection is involved in context binding. This result is well in line with previous studies suggesting that the processes underlying context memory and R responses may be largely identical. For example, the rate of R responses rises with the distinctiveness of sources, which in turn enhances memory for context (Conway & Dewhurst, 1995; Donaldson, MacKenzie, & Underhill, 1996). Furthermore, activity in the same brain areas, such as the hippocampus and prefrontal regions, appears to be characteristic of both context memory and R responses, (e.g., Eldridge, Knowlton, Furmanski, Bookheimer, & Engel, 2000; Yonelinas, Otten, Shaw, & Rugg, 2005). Contrary evidence, however, can be found in studies in which R responses and context memory were dissociated (e.g., Hicks et al., 2002; Meiser & Sattler, 2007). Thus the conclusion that both context memory and recollection are the product of the same process may be premature.

Furthermore, other DT manipulations might be more successful in dissociating context–context binding and the recollective experience. For instance, secondary tasks which tap visuo-spatial processing as opposed to purely central processing might be more effective in impairing binding of visuo-spatial features. On the other hand, the evidence for resource-demanding processing in feature binding in visual short-term memory is not unequivocal (see Baddeley, Allen, & Hitch, 2011, for a review). Thus there might be moderators for the role of effortful processing in feature binding. One likely candidate is whether the features are stimulus-bound or spatio–temporal. Evidence from differences in the processing demands of stimulus-bound and spatio-temporal features comes from earlier studies (Spencer & Raz, 1995; Troyer et al., 1999) but can also be found here. In the second experiment, in which context memory was allowed to vary between groups, only memory for position on the screen but not memory for font size was affected by the DT load. Interestingly, concepts similar to the distinction between stimulus-bound and spatial temporal features have been proposed in the visual cognition literature under the rubric of unitization effects (Ecker, Maybery, & Zimmer, 2013; Walker & Cuthbert, 1998). The context features used here could be seen as a combination of stimulus-bound (i.e., font size) and spatio-temporal (i.e., position on the screen) and it therefore remains an interesting question for future research whether binding of combinations of purely spatio-temporal context features are more prone to dual-task manipulations. In line with the distinction between stimulus-bound and spatio-temporal context features the present results can be taken as an indication that different context features may also differ in their role for the subjective retrieval experience. While memory for the rather stimulus-bound feature font size was not prone to the DT manipulation, memory for the spatial temporal feature position was reduced by DT load and this effect was parallel to the effect of DT load on recollective experience. Future research should thus investigate whether differences in subjective retrieval experience could be related to differences in the processing of different context details.

The evidence for the binding of context features in episodic memory, however, goes beyond the demonstration of visual feature binding. Previous research has shown that conscious recollection is associated with cross-modal binding. Meiser and colleagues (Meiser & Sattler, 2007; Meiser et al., 2008), for instance, used visual context features such as font size and an auditory semantic feature such as gender of a speaker to show that binding of these features too was characteristic of conscious recollection. Whereas it is possible that visuo-spatial features might be bound in rather early stages of processing (see Luck & Vogel, 1997), features of different modalities might be more likely to rely on a central binding process. This idea receives preliminary support by a study showing that prefrontal activity did not differ between a condition in which two visual features were part of the same object and a condition in which two visual features were spatially separated (Kuo & Van Petten, 2008). Parietal activity, however, was correlated with context memory performance, indicating that visual processing was mainly involved in binding of these visual features. Future research should thus extend the investigation to other combinations of context features.

7. Conclusions

Regarding the role of overall context memory for the recollective experience, the present research offers original evidence for the causal role of context memory in producing the subjective experience of conscious recollection by looking at DT effects on the subjective retrieval experience and context memory jointly. In Experiment 1 experimentally equated similar levels of overall context memory in the DT group and the control group were accompanied by similar levels of recollective experience as indicated by model-based analyses. In Experiment 2, on the other hand, differences in overall context memory between the DT during encoding group and the control group were accompanied by significant differences in the recollective experience such that lower levels of context memory went together with lower levels of recollective experience. Importantly, however, the general relation of higher context memory for *R* than for *K* responses, however, was preserved across both experiments as well as under cognitive load during encoding or retrieval.

While cognitive load during encoding leads to reductions in the frequency of the subjective experience of conscious recollection and reductions in overall context memory, context–context binding was *not* affected over and above the effects on conscious recollection. Thus, given participants report recollective retrieval experiences, there was no difference in the quality of the associated memory characteristics in terms of context memory and context–context binding.

References

Allen, R. J., Baddeley, A. D., & Hitch, G. J. (2006). Is the binding of visual features in working memory resource-demanding? *Journal of Experimental Psychology: General, 135*, 298–313.

Baddeley, A. D., Allen, R. J., & Hitch, G. J. (2011). Binding in visual working memory: The role of the episodic buffer. *Neuropsychologia, 49*, 1393–1400.

Batchelder, W. H., & Riefer, D. M. (1990). Multinomial processing models of source monitoring. *Psychological Review, 97*, 548–564.

Bayen, U. J., Murnane, K., & Erdfelder, E. (1996). Source discrimination, item detection, and multinomial models of source monitoring. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 22*, 197–215.

Boywitt, C. D., Kuhlmann, B. G., & Meiser, T. (2012). The role of source memory in older adults' recollective experience. *Psychology and Aging, 27*, 484–497.

Boywitt, C. D., & Meiser, T. (2012a). Bound context features are integrated at encoding. *The Quarterly Journal of Experimental Psychology*, 1484–1501.

Boywitt, C. D., & Meiser, T. (2012b). The role of attention for context–context binding of intrinsic and extrinsic features. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 38*, 1099–1107.

Brown, L. A., & Brockmole, J. R. (2010). The role of attention in binding visual features in working memory: Evidence from cognitive ageing. *The Quarterly Journal of Experimental Psychology, 63*, 2067–2079.

Conway, M. A., & Dewhurst, S. A. (1995). Remembering, familiarity, and source monitoring. *The Quarterly Journal of Experimental Psychology, 48*, 125–140.

Craik, F. I. M., Luo, L., & Sakuta, Y. (2010). Effects of aging and divided attention on memory for items and their contexts. *Psychology and Aging, 25*, 968–979.

Delvenne, J. F., Cleeremans, A., & Laloyaux, C. (2010). Feature bindings are maintained in visual short-term memory without sustained focused attention. *Experimental Psychology, 57*, 108–116.

Dewhurst, S. A., & Hitch, G. J. (1999). Cognitive effort and recollective experience in recognition memory. *Memory, 7*, 129–146.

Donaldson, W., MacKenzie, T. M., & Underhill, C. F. (1996). A comparison of recollective memory and source monitoring. *Psychonomic Bulletin & Review, 3*, 486–490.

Dudukovic, N. M., & Knowlton, B. J. (2006). Remember-know judgments and retrieval of contextual details. *Acta Psychologica, 122*, 160–173.

Ecker, U. K. H., Maybery, M., & Zimmer, H. D. (2013). Binding of intrinsic and extrinsic features in working memory. *Journal of Experimental Psychology: General, 142*, 218–234.

Ecker, U. K. H., Zimmer, H. D., & Groh-Bordin, C. (2007). Color and context: An ERP study on intrinsic and extrinsic feature binding in episodic memory. *Memory & Cognition, 35*, 1483–1501.

Eldridge, L. L., Knowlton, B. J., Furmanski, C. S., Bookheimer, S. Y., & Engel, S. A. (2000). Remembering episodes: A selective role for the hippocampus during retrieval. *Nature Neuroscience, 3*, 1149–1152.

Elsley, J. V., & Parmentier, F. B. R. (2009). Is verbal–spatial binding in working memory impaired by a concurrent memory load? *The Quarterly Journal of Experimental Psychology, 62*, 1696–1705.

Erdfelder, E., Auer, T. S., Hilbig, B. E., Assfalg, A., Moshagen, M., & Nadarevic, L. (2009). Multinomial processing tree models a review of the literature. *Zeitschrift für Psychologie/Journal of Psychology, 217*, 108–124.

Gardiner, J. M., Gregg, V. H., Mashru, R., & Thaman, M. (2001). Impact of encoding depth on awareness of perceptual effects in recognition memory. *Memory & Cognition, 29*, 433–440.

Gardiner, J. M., & Parkin, A. J. (1990). Attention and recollective experience in recognition memory. *Memory & Cognition, 18*, 579–583.

Hicks, J. L., Marsh, R. L., & Ritschel, L. (2002). The role of recollection and partial information in source monitoring. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 28*, 503–508.

Jacoby, L. L. (1991). A process dissociation framework – Separating automatic from intentional uses of memory. *Journal of Memory and Language, 30*, 513–541.

Kuo, T. Y., & Van Petten, C. (2008). Perceptual difficulty in source memory encoding and retrieval: Prefrontal versus parietal electrical brain activity. *Neuropsychologia, 46*, 2243–2257.

Light, L. L., & Berger, D. E. (1976). Are there long-term literal copies of visually presented words. *Journal of Experimental Psychology: Human Learning and Memory, 2*, 654–662.
Luck, S. J., & Vogel, E. K. (1997). The capacity of visual working memory for features and conjunctions. *Nature, 390*, 279–281.
Mandler, G. (1980). Recognizing – The judgment of previous occurrence. *Psychological Review, 87*, 252–271.
Meiser, T. (2009, October). Multinomial analysis of multidimensional source memory: Overview and new extensions. *Paper presented at the Winer Lectures*, Purdue University, West Lafayette.
Meiser, T., & Bröder, A. (2002). Memory for multidimensional source information. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 28*, 116–137.
Meiser, T., & Sattler, C. (2007). Boundaries of the relation between conscious recollection and source memory for perceptual details. *Consciousness and Cognition, 16*, 189–210.
Meiser, T., Sattler, C., & Weisser, K. (2008). Binding of multidimensional context information as a distinctive characteristic of remember judgments. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 34*, 32–49.
Moscovitch, M. (1992). Memory and working-with-memory – A component process model based on modules and central systems. *Journal of Cognitive Neuroscience, 4*, 257–267.
Moshagen, M. (2010). Multitree: A computer program for the analysis of multinomial processing tree models. *Behavior Research Methods, 42*, 42–54.
Mulligan, N. W. (2011). Generation disrupts memory for intrinsic context but not extrinsic context. *The Quarterly Journal of Experimental Psychology, 64*, 1543–1562.
Murnane, K., & Bayen, U. J. (1996). An evaluation of empirical measures of source identification. *Memory & Cognition, 24*, 417–428.
Parkin, A. J., & Russo, R. (1990). Implicit and explicit memory and the automatic effortful distinction. *European Journal of Cognitive Psychology, 2*, 71–80.
Parkin, A. J., & Walter, B. M. (1992). Recollective experience, normal aging, and frontal dysfunction. *Psychology and Aging, 7*, 290–298.
Perfect, T. J., Mayes, A. R., Downes, J. J., & VanEijk, R. (1996). Does context discriminate recollection from familiarity in recognition memory? *The Quarterly Journal of Experimental Psychology, 49*, 797–813.
Rajaram, S. (1993). Remembering and knowing – Two means of access to the personal past. *Memory & Cognition, 21*, 89–102.
Rensink, R. A. (2000). The dynamic representation of scenes. *Visual Cognition, 7*, 17–42.
Riefer, D. M., Chien, Y., & Reimer, J. F. (2007). Positive and negative generation effects in source monitoring. *The Quarterly Journal of Experimental Psychology, 60*, 1389–1405.
Spencer, W. D., & Raz, N. (1995). Differential effects of aging on memory for content and context: A meta-analysis. *Psychology and Aging, 10*, 527–539.
Starns, J. J., & Hicks, J. L. (2008). Context attributes in memory are bound to item information, but not to one another. *Psychonomic Bulletin & Review, 15*, 309–314.
Stefurak, D. L., & Boynton, R. M. (1986). Independence of memory for categorically different colors and shapes. *Perception & Psychophysics, 39*, 164–174.
Treisman, A., & Gelade, G. (1980). Feature-integration theory of attention. *Cognitive Psychology, 12*, 97–136.
Treisman, A., & Schmidt, H. (1982). Illusory conjunctions in the perception of objects. *Cognitive Psychology, 14*, 107–141.
Troyer, A. K., & Craik, F. I. M. (2000). The effect of divided attention on memory for items and their context. *Canadian Journal of Experimental Psychology, 54*, 161–171.
Troyer, A. K., Winocur, G., Craik, F. I. M., & Moscovitch, M. (1999). Source memory and divided attention: Reciprocal costs to primary and secondary tasks. *Neuropsychology, 13*, 467–474.
Tulving, E. (1985). Memory and consciousness. *Canadian Psychology, 26*, 1–12.
Uncapher, M. R., Otten, L. J., & Rugg, M. D. (2006). Episodic encoding is more than the sum of its parts: An fmri investigation of multifeatural contextual encoding. *Neuron, 52*, 547–556.
Walker, P., & Cuthbert, L. (1998). Remembering visual feature conjunctions: Visual memory for shape-colour associations is object-based. *Visual Cognition, 5*, 409–455.
Wheeler, M. E., & Treisman, A. M. (2002). Binding in short-term visual memory. *Journal of Experimental Psychology: General, 131*, 48–64.
Yonelinas, A. P. (1994). Receiver-operating characteristics in recognition memory – Evidence for a dual-process model. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 20*, 1341–1354.
Yonelinas, A. P. (2001). Consciousness, control, and confidence: The 3 Cs of recognition memory. *Journal of Experimental Psychology-General, 130*, 361–379.
Yonelinas, A. P. (2002). The nature of recollection and familiarity: A review of 30 years of research. *Journal of Memory and Language, 46*, 441–517.
Yonelinas, A. P., Otten, L. J., Shaw, K. N., & Rugg, M. D. (2005). Separating the brain regions involved in recollection and familiarity in recognition memory. *Journal of Neuroscience, 25*, 3002–3008.